Final IEEE Conference Presentation Script

Slide 1 — Title Slide

Good morning respected judges, professors, and fellow participants.

My name is **Dharun R**, and today I am presenting our research titled:

“Correlation Mapping of Sanitation Access and Public Health Outcomes in Tamil Nadu.”

This work was carried out in collaboration with my team from the **Department of Artificial Intelligence and Data Science at SRM Madurai College for Engineering and Technology.**

The goal of this research is to analyze how **sanitation infrastructure influences public health outcomes across districts in Tamil Nadu** using statistical and spatial analysis techniques.

Slide 2 — Introduction

Sanitation is a fundamental determinant of public health and social well-being.

In many developing regions, inadequate sanitation leads to serious challenges such as:

- Waterborne diseases
- Poor environmental hygiene
- Increased pressure on healthcare systems

Although Tamil Nadu has made significant progress in sanitation infrastructure through national programs like the **Swachh Bharat Mission**, disparities still exist across districts.

Understanding how sanitation access relates to public health outcomes can help policymakers design **more effective sanitation and healthcare strategies.**

Slide 3 — Problem Statement

Despite improvements in sanitation facilities, many regions still experience poor health outcomes.

The major challenges include:

- Unequal access to sanitation facilities across districts

- Inefficient waste management systems
- Lack of data-driven policy decisions in sanitation planning

Therefore, there is a strong need to analyze the **relationship between sanitation access and public health indicators using statistical methods and spatial analysis.**

Slide 4 — Research Objectives

The objectives of this study are:

1. To analyze sanitation accessibility across districts in Tamil Nadu
2. To evaluate the relationship between sanitation access and public health outcomes
3. To identify patterns using correlation analysis and spatial mapping
4. To provide insights that can support improved sanitation policies and health planning

Through this research, we aim to contribute to **evidence-based public health decision making.**

Slide 5 — Dataset and Study Area

This study focuses on district-level data from **Tamil Nadu**.

Two major categories of variables were analyzed.

The first category is **sanitation indicators**, which include:

- Household toilet access
- Waste disposal facilities
- Open defecation rates

The second category is **health indicators**, which include:

- Infant mortality rates
- Incidence of waterborne diseases such as diarrhea and cholera

The datasets were collected from multiple sources including **government public health records, sanitation statistics, and national survey databases.**

Slide 6 — Methodology

The study follows a **data-driven analytical framework**.

The methodology consists of four main steps:

First, collecting sanitation and health data from official sources.

Second, cleaning and preprocessing the dataset to remove inconsistencies.

Third, applying statistical analysis techniques such as correlation and regression.

Finally, visualizing the results using GIS mapping and data visualization tools.

For implementation, tools such as **Python, statistical analysis methods, and visualization techniques** were used.

Slide 7 — Correlation Analysis

To measure the relationship between sanitation indicators and health outcomes, we used the **Pearson Correlation Coefficient**.
formula:

$$r = \frac{\Sigma\, (xi - \bar{x})(yi - \bar{y})}{\sqrt{[\, \Sigma\, (xi - \bar{x})2\ \Sigma\, (yi - \bar{y})2\,]}}$$

In this formula:

- $x_i$represents sanitation indicator values
- $y_i$represents health outcome values
- $\bar{x}$and $\bar{y}$represent the mean values

The value of **r** helps interpret the strength of the relationship:

- **+1 indicates a strong positive correlation**
- **0 indicates no correlation**
- **−1 indicates a strong negative correlation**

This analysis helps identify whether improved sanitation access leads to better public health outcomes.

Slide 8 — Data Visualization

To better understand the patterns in the dataset, we used several visualization techniques.

These include:

- Correlation heatmaps
- Scatter plots
- District-level comparison graphs
- GIS-based spatial mapping

These visual tools help identify relationships between sanitation availability and disease prevalence and also highlight **districts that may require targeted intervention.**

Slide 9 — Future Work

Future research can expand this study in several directions.

Possible improvements include:

- Extending the analysis to other states in India
- Applying machine learning models to predict health risks
- Studying long-term sanitation trends
- Including environmental factors such as water quality

These improvements can help develop **more accurate public health prediction models.**

Slide 10 — Conclusion

In conclusion, this study examined the relationship between sanitation access and public health outcomes in Tamil Nadu.

The key findings are:

- Sanitation access has a measurable impact on health outcomes
- Correlation analysis helps identify sanitation gaps across districts

- Improving sanitation infrastructure can significantly enhance public health conditions

Overall, strengthening sanitation systems and improving hygiene awareness can contribute to **healthier and more sustainable communities.**

Slide 11 — Thank You

Thank you for your time and attention.

I would be happy to answer any questions or discuss the research further.